

LINCOLN, NE, USA

WMO: 725510

Lat: 40.851N

Lon: 96.748W

Elev: 1190

StdP: 14.07

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14939

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.4	2.2	-10.4	3.3	-1.0	-6.1	4.1	3.8	31.6	27.3	27.9	26.5	6.8	320	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	96.3	75.2	93.2	74.8	90.4	73.7	78.6	90.8	77.2	89.2	75.5	86.7	13.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	137.9	86.3	73.5	130.2	84.5	72.0	123.7	82.8	43.1	91.0	41.4	89.4	39.9	87.2	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.4	24.6	21.4	-11.3	101.7	4.6	2.8	-14.5	103.7	-17.2	105.3	-19.8	106.9	-23.1	108.9	
			DB												
			WB	-11.7	81.1	4.4	1.4	-14.9	82.2	-17.5	83.0	-20.0	83.8	-23.2	84.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	24.8	28.8	40.7	51.8	62.8	73.2	77.8	75.3	67.2	53.8	40.1	28.3
	DBStd	20.86	11.95	12.31	12.02	9.80	8.59	6.76	6.00	5.85	8.47	9.62	10.03	10.85
	HDD50	2881	781	597	335	94	6	0	0	0	2	69	322	675
	HDD65	5913	1246	1013	756	408	148	13	1	3	74	365	747	1139
	CDD50	3675	0	4	46	148	402	697	863	784	518	187	25	1
	CDD65	1230	0	0	2	11	78	260	399	322	140	18	0	0
	CDH74	13441	0	1	37	201	898	2780	4514	3253	1524	223	10	0
	CDH80	5743	0	0	6	54	303	1149	2154	1432	585	59	1	0
Wind	WSAvg	9.7	9.3	10.0	11.0	12.0	10.4	9.8	8.6	8.2	9.1	9.6	9.6	9.0
Precipitation	PrecAvg	28.8	0.6	0.9	2.0	2.8	4.1	4.1	3.0	3.5	3.5	2.1	1.3	0.8
	PrecMax	41.3	1.6	3.1	6.7	7.2	10.2	12.9	6.9	8.6	8.3	5.4	3.8	3.4
	PrecMin	17.9	0.0	0.1	0.1	0.3	0.9	0.2	0.4	0.1	0.3	0.0	0.0	0.0
	PrecStd	6.1	0.4	0.7	1.6	1.7	1.9	2.9	1.5	2.2	2.2	1.6	1.1	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.6	68.5	78.3	86.4	92.4	96.9	101.4	99.2	94.5	87.2	74.4	60.4
		MCWB	45.3	51.1	58.7	63.4	69.8	74.6	75.8	74.8	72.4	67.4	57.5	49.4
	2%	DB	51.7	59.1	72.4	79.7	87.2	93.0	97.2	94.9	90.6	79.6	67.6	54.2
		MCWB	42.0	47.6	56.4	61.2	68.3	74.1	76.1	75.6	72.0	63.3	54.2	45.2
	5%	DB	46.4	52.7	66.2	74.4	83.1	89.9	93.8	91.3	86.8	75.2	61.9	49.2
		MCWB	39.4	43.6	53.5	58.8	65.8	73.2	76.0	74.8	70.3	61.2	51.3	41.4
	10%	DB	41.4	46.7	60.3	69.5	79.0	86.8	90.5	87.9	82.8	70.6	57.0	44.2
		MCWB	36.3	40.0	50.3	55.9	64.7	71.4	75.1	73.4	68.2	58.8	48.2	38.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.5	54.6	63.3	66.5	73.8	79.1	81.1	80.2	76.0	70.6	60.3	53.1
		MCDB	56.8	61.0	73.6	79.2	86.3	90.9	93.3	92.0	88.7	81.6	68.0	58.1
	2%	WB	42.9	48.3	59.3	63.4	71.2	76.7	79.2	78.1	74.0	66.9	56.6	45.9
		MCDB	50.7	58.8	66.2	75.6	82.6	88.8	91.8	90.0	85.8	75.3	65.0	52.7
	5%	WB	39.6	43.9	54.9	60.4	68.8	75.0	77.7	76.4	72.4	63.6	52.6	41.9
		MCDB	46.0	51.9	65.7	71.8	79.7	86.6	90.1	87.7	83.5	72.3	60.9	48.6
	10%	WB	36.6	40.2	50.4	57.4	66.2	73.3	76.2	74.8	70.5	60.0	48.4	38.5
		MCDB	41.3	46.6	60.0	68.0	76.0	84.5	87.8	85.0	80.3	68.4	56.5	43.7
Mean Daily Temperature Range	5% DB	MDBR	19.9	20.4	23.4	23.6	22.3	21.3	21.3	21.4	24.2	23.7	22.5	19.7
		MCDBR	28.5	30.4	32.9	31.9	28.0	24.3	25.0	25.5	27.1	30.1	31.1	28.1
		MCWBR	19.8	19.7	18.9	16.9	12.8	10.4	9.4	10.2	11.7	15.7	19.0	19.8
	5% WB	MCDBR	26.9	28.9	28.5	27.4	23.7	21.4	21.5	22.1	23.2	23.7	27.9	26.8
		MCWBR	19.4	19.2	17.1	16.2	12.5	10.8	9.7	10.4	11.3	14.0	18.5	19.7
Clear-Sky Solar Irradiance	taub	0.283	0.303	0.326	0.381	0.407	0.418	0.424	0.422	0.382	0.333	0.295	0.281	
	taud	2.468	2.430	2.408	2.276	2.271	2.302	2.283	2.294	2.353	2.478	2.509	2.478	
	Ebn at Noon	281	290	294	283	276	273	270	267	271	275	274	272	
	Edh at Noon	24	29	33	41	42	41	41	40	35	28	23	22	
All-Sky Solar Radiation	RadAvg	670	954	1249	1587	1864	2091	2083	1813	1485	1059	748	564	
	RadStd	58	58	106	104	137	142	128	101	97	115	52	55	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page